

EXPERIMENT 1

1. AIM:

To acquire data from multiple sources and perform data profiling using Microsoft Power BI in order to assess data quality, understand data structure, identify anomalies, and prepare data for further analytical processing.

2. OBJECTIVES:

- To collect and integrate data from multiple sources using Microsoft power BI.
- To acquire data from multiple heterogeneous sources such as CSV, Excel, and SQL Server.
- To import and integrate different data sets into Microsoft Power BI.
- To assess and improve the overall quality of the data.

3. TOOLS / SOFTWARE REQUIRED:

- Microsoft Power BI Desktop
- Sample datasets: Sales_Data.csv, Employee_DB.xlsx, and a sample SQL database

4. DATASET USED: Sales_Data.csv, Employee_DB.xlsx, Transactions.sql

5. THEORY:

- Business Intelligence (BI) refers to the technologies and processes used to collect, integrate, analyze, and present business data to generate meaningful insights and support better decision-making. Key components of BI include:
 - Data Sources: Databases, flat files, cloud storage, and APIs.
 - ETL Pipeline: Extracts, transforms, and loads data for processing.
 - Data Warehouse/Data Lake: Stores structured or unstructured data.
 - BI & Reporting Tools: Tools such as Microsoft Power BI for data transformation and visualization.
 - Analytics Layer: Supports descriptive, diagnostic, predictive, and prescriptive analytics.
- Data Acquisition is the process of collecting and importing data from different sources into a common environment for processing. In Power BI, data can be acquired using the Get Data option from sources such as CSV, Excel, SQL Server, cloud services, and web-based sources.
- Data Profiling involves examining a dataset to understand its structure, content, and quality before further analysis. It helps assess important data quality dimensions such as completeness, uniqueness, validity, consistency, accuracy, and timeliness.
- Data Profiling in Power BI – Power Query Editor: Power BI provides built-in data profiling features through Power Query Editor:
 - Column Quality: Shows the percentage of valid, empty, and error values.
 - Column Distribution: Displays value frequency along with distinct and unique counts.
 - Column Profile: Provides detailed statistics such as minimum, maximum, mean, median, and standard deviation for selected columns.

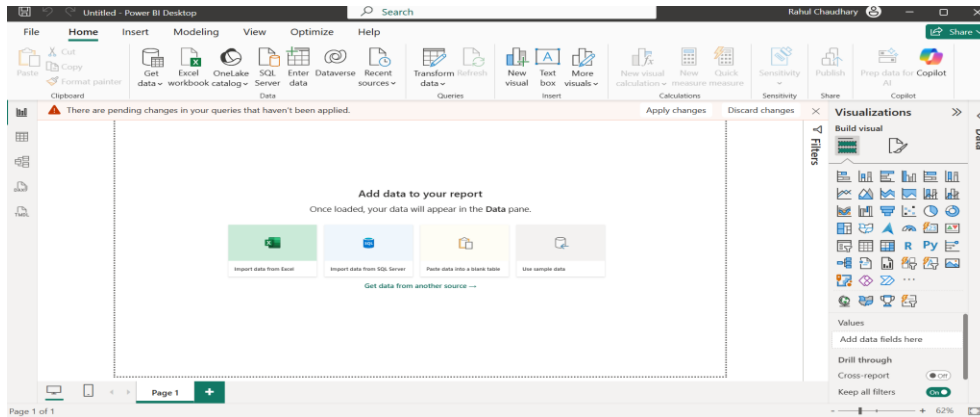
6. PROCEDURE:

Part A: Software Setup and Data Acquisition

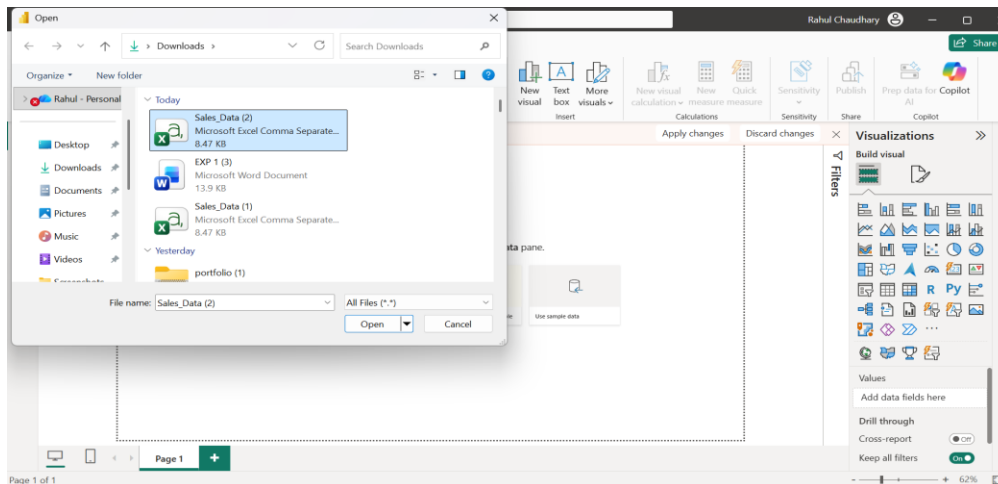
Step 1: Install the required software.

- Download and install Microsoft Power BI Desktop.

Step 2: Launch Power BI Desktop and import the CSV file.



Home > Get Data > Text/CSV

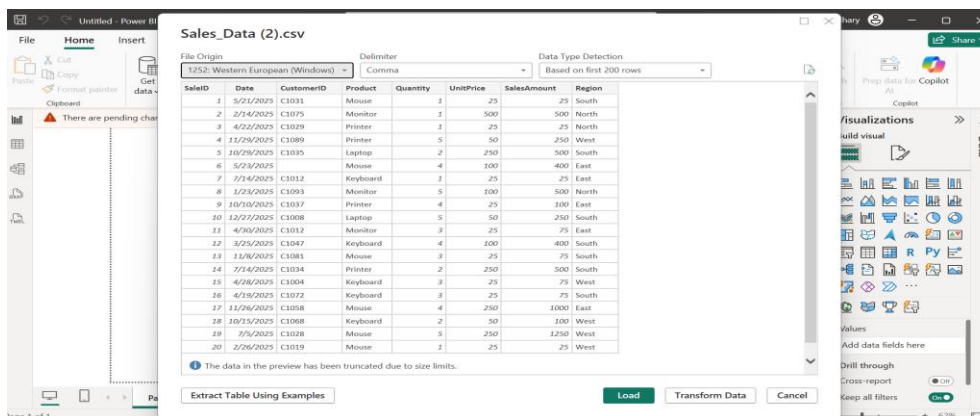


Browse to Sales_Data.csv > Load

Part B: Data Profiling in Power Query Editor

Step 3: Open Power Query Editor.

Home > Transform Data



Step 4: Enable data profiling features.

View tab > Tick Column Quality

The screenshot shows the Power Query Editor interface. The 'View' tab is selected, and the 'Column quality' checkbox is checked. The data preview shows a table with columns: SaleID, Date, CustomerID, Product, and Quantity. The 'Query Settings' pane on the right shows the 'Properties' tab with 'Sales_Data (2)' as the source and 'Promoted Headers' and 'Changed Type' as applied steps.

View tab > Tick Column Distribution

The screenshot shows the Power Query Editor interface. The 'View' tab is selected, and the 'Column distribution' checkbox is checked. The data preview shows the same table as before, but with additional distribution charts for each column. The 'Query Settings' pane on the right shows the 'Properties' tab with 'Sales_Data (2)' as the source and 'Promoted Headers' and 'Changed Type' as applied steps.

View tab > Tick Column Profile

The screenshot shows the Power Query Editor interface. The 'View' tab is selected, and the 'Column profile' checkbox is checked. The data preview shows the same table as before, but with additional column statistics and value distribution charts. The 'Query Settings' pane on the right shows the 'Properties' tab with 'Sales_Data (2)' as the source and 'Promoted Headers' and 'Changed Type' as applied steps.

Column statistics

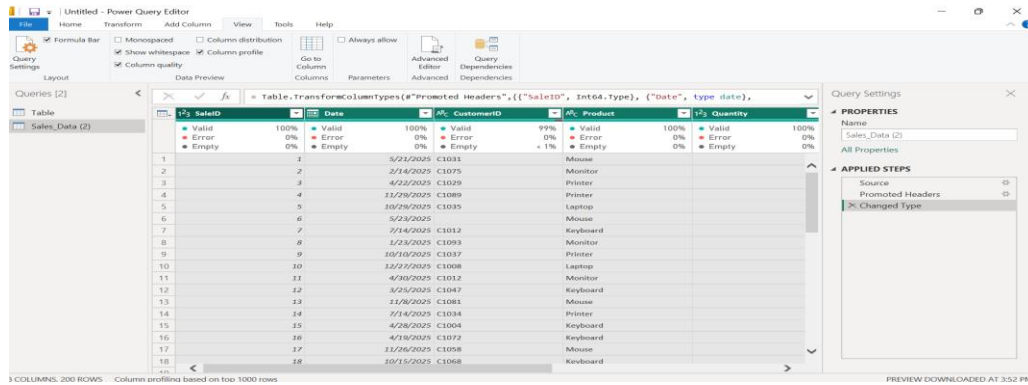
Statistic	Value
Distinct	200
Unique	200
NaN	0
Zero	0
Min	1
Max	200
Average	100.5
Standard deviation	57.8791...
Even	100
Odd	100

Value distribution

Quantity: 5 (2%)
Distinct: 0 (0%)
Unique: 0 (0%)

Remove Duplicates

Select each column one at a time and observe the profiling statistics displayed below the data preview grid.



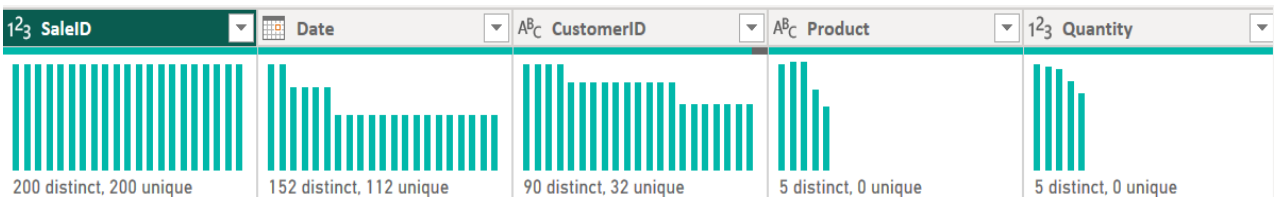
Step 5: Record profiling observations.

For each column in each dataset, observe and record:

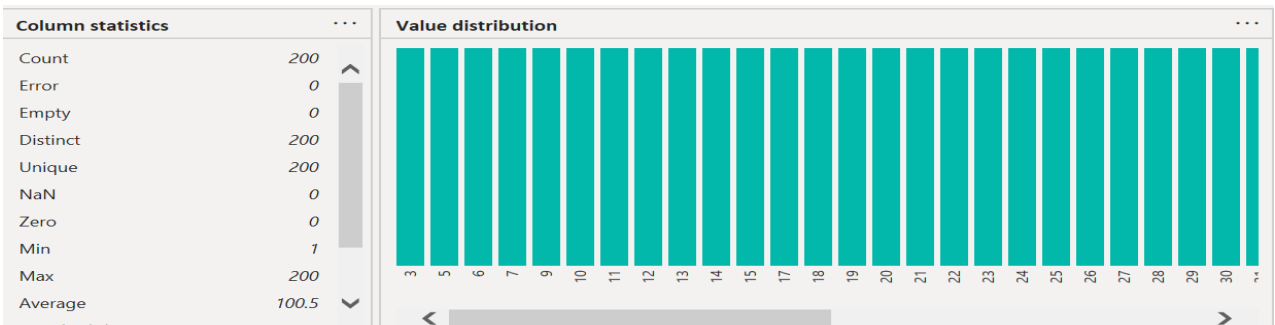
- Data type and total row count.
- Null/empty percentage from Column Quality.

1 ² 3 SaleID	Date	A ^B C CustomerID	A ^B C Product	1 ² 3 Quantity
Valid 100%	Valid 100%	Valid 99%	Valid 100%	Valid 100%
Error 0%	Error 0%	Error 0%	Error 0%	Error 0%
Empty 0%	Empty 0%	Empty < 1%	Empty 0%	Empty 0%

- Distinct and unique value counts from Column Distribution.



- Minimum, maximum, mean, and standard deviation for numeric columns from Column Profile.
- Any errors, mismatched data types, or unexpected values.

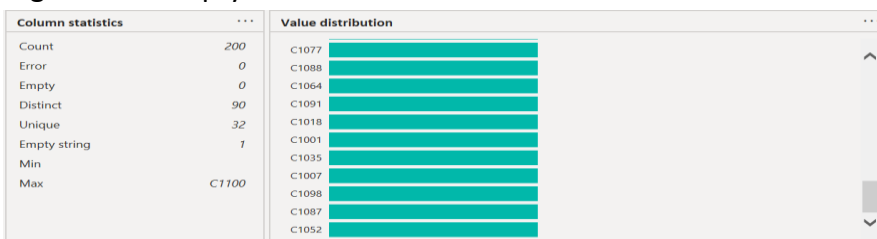


Part C: Identify and Document Data Quality Issues

Step 6: Identify data quality issues from the profiling results.

Check for common issues such as:

- High null or empty values in customer id columns.



- Columns containing only one unique value.
- ➔ There are no any columns containing one unique value.
- Numeric columns containing text values or incorrect data types.
- Date values outside the expected range or future dates.
- Duplicate records indicated by low uniqueness.
- ➔ Duplicate rows were identified in the dataset. The selected identifier column contains 200 rows, 90 distinct values, and 0 unique values, indicating substantial repetition. Duplicate records should be reviewed and removed or consolidated where appropriate.

Step 7: Document the identified issues.

Record at least three data quality issues found during profiling and mention a suitable remediation action for each issue.

8.RESULT

Data from multiple heterogeneous sources (CSV, Excel, and SQL database) was successfully acquired and imported into Microsoft Power BI Desktop. Data profiling was performed using the Power Query Editor's built-in tools — Column Quality, Column Distribution, and Column Profile. Key data quality dimensions including completeness, uniqueness, and validity were assessed across all datasets, and data quality issues were identified and documented for remediation prior to analysis.

9.LEARNING OUTCOMES

- Understood the concept of Business Intelligence and its role in data-driven decision-making.
- Learned how to acquire data from different heterogeneous sources.
- Learned how to import CSV, Excel, and SQL Server data into Power BI.
- Gained practical experience with Microsoft Power BI Desktop.
- Learned how to use SQL Server Management Studio for working with SQL-based data.
- Learned how to use Power Query Editor for data profiling.
- Understood the purpose of Column Quality, Column Distribution, and Column Profile.
- Learned how to identify missing values, duplicate records, data type mismatches, and other data quality issues.
- Understood the importance of data profiling before performing further analysis and visualization.
- Gained practical experience in creating a basic BI dashboard using Power BI.